Reading Shallow Flood Depth from Real USGS Gauge-Camera Footage with a Learned Staff-Plate Reader

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Abstract

How accurately can a fixed camera estimate flood water depth, and how does accuracy change with image quality and camera resolution? I study one real USGS camera watching a staff plate in shallow overland flow: 2259 frames over 15 months (2023–2026), each labelled with the official gauge height. A learned reader registers each frame, rectifies the plate, classifies plate rows as submerged, and decodes the water level; an image-quality gate abstains on bad frames. Evaluated by leaving out whole months, it reaches 1.13 cm mean absolute error on the 67% of frames that pass the gate (6.97 cm for a colour-threshold baseline) and 0.79 cm on two locked months; without the gate the errors are 3.38 and 3.11 cm. Error stays below twice its native value up to about 5.0 cm per pixel. Limits: one camera, and water below the plate bottom (4.3 cm) is unobservable.

1 Problem and Motivation

Question. Given one frame from a fixed camera that sees a reference object of known scale, estimate the water depth h (cm) at the object. Any flood decision (an alarm threshold, severity classes) is a function of h, so I evaluate h itself and report error by depth range rather than fixing a threshold. One row is one frame at one camera and time. The prediction uses the current frame, a one-time setup (reference views, plate geometry) and, for pose recovery only, earlier frames; the gauge is never available at prediction time. Camera water-level work mostly targets rivers or deep urban water (Vandaele et al., 2021; Liang et al., 2023); this camera sees shallow overland flow. The data are one camera (USGS 411229084541101, overland flow near Harlan, Indiana), so claims are limited to generalisation across months of that camera (Sec. 7).

2 Data and Preparation

Source. Frames come from the public USGS National Imagery Management System bucket (USGS, 2026); labels are the USGS gage height, both public domain. NWIS was not reachable from my environment, so the gauge values come from the committed result tables of two open-source projects (waterline & stagesight, 2026) (not re-checked against NWIS); no camera-derived value is a label. The plate spans 0–62 cm; the gauge sits beside it.

Content. 2259 frames (median spacing 21 min) in 15 month blocks; gauge -0.6 to 55 cm; 35% dry (≤ 0.5 cm), 25% infrared night. The camera pose changes between months.

Quality checks. SIFT registration (Lowe, 2004) to a reference view succeeds on 82% of frames (the rest reuse the last good pose); a one-frame-per-month montage and the failure frames show fog, mud

Table 1: Held-out depth error (cm). Gated = frames passing the quality gate (coverage 67% CV, 82% locked); bias = mean(estimate − gauge); last three columns: MAE by gauge depth range; CI = block bootstrap.

Eval.	Frames	Method	n	MAE [95% CI]	RMSE	bias	dry	0.5–10	>10
CV	gated	learned reader	1319	1.13 [0.7, 1.7]	3.1	+0.4	0.4	2.0	1.5
	gated	threshold baseline	1319	6.97 [6.0, 8.4]	10.1	+1.1	5.9	8.0	7.8
	gated	always 0 cm	1319	8.80 [5.0, 14.8]	12.9	-8.7	0.0	6.9	18.3
	all	learned reader	1976	3.38 [2.1, 4.5]	7.3	+0.5	1.9	5.0	4.1
	all	threshold baseline	1976	7.94 [7.4, 8.7]	10.9	+2.5	8.0	9.4	7.6
Locked	gated	learned reader	233	0.79	1.2	+0.0	0.0	1.6	0.6
	gated	threshold baseline	233	11.52	14.6	-8.7	9.2	7.2	12.5
	gated	always 0 cm	233	18.90	21.1	-18.9	0.0	7.6	21.9
	all	learned reader	283	3.11	8.9	-2.1	1.6	2.1	3.4
	all	threshold baseline	283	11.67	14.8	-8.1	11.7	7.8	12.4

or droplets on the lens, glare and over-exposure; the plate bottom is at 4.3 cm, so shallower water cannot be observed on it.

Preprocessing and leakage control. Frames are resized to 1280 px; the plate quad is rectified to a 360×48 strip (row ≈ 0.16 cm). Everything learned is fitted on training months only: the dry template (per-row median Lab colour of dry daytime frames), the affine calibration, the classifier, the baseline threshold and the gate thresholds (training Laplacian-variance quantiles). Registration reference views come from development months and use geometry, never labels.

3 Methods

Baselines. B0: always 0 cm (a dry scene). B1: a row is wet if its colour change from the dry template exceeds τ (tuned on training months), then the same decoding and calibration as below.

Learned reader. For frame i and strip row k (height h_k), a gradient-boosted classifier (scikit-learn `HistGradientBoosting`, Pedregosa et al., 2011) outputs $p_{ik} = P(\text{row submerged})$ from row colour, change from the template, texture, vertical context and frame quality. Labels are $y_{ik} = \mathbb{1}[h_k < g_i]$ (g_i the gauge height): no manual annotation. Water level is the monotone change point $b^\star = \arg\max_b \sum_{k<b} \log(1-p_{ik}) + \sum_{k\geq b} \log p_{ik}$, mapped to height and then to centimetres by a least-squares affine fit; levels at or below the plate bottom report 0. The log-likelihood gap to the runner-up boundary is the confidence *margin*.

Quality gate. A frame is rejected when the Laplacian variance v of the whole image lies outside $[0.55, 1.7]$ times the training median (low: fog, blur, obstruction; high: lens dirt, droplets, glare).

Error model. With plate scale s (cm/px) and waterline error σ_v (px), $\mathrm{Var}(\hat{h}) = s^2\sigma_v^2 + \sigma_g^2$; σ_g collects gauge–plate offset and calibration error (quantisation adds $s^2/12$).

4 Evaluation Design

The split unit is the calendar-month block, never the frame: consecutive frames are near duplicates and lighting and pose are shared within a month. Two blocks (2025-04, 2026-04) were locked in `make_usgs_splits.py` before any model was trained; the other 13 (1976 frames) support leave-one-block-out cross-validation (CV) and all tuning (classifier hyperparameters were left at fixed defaults). **Disclosure:** I looked at the locked blocks twice. The first gate was one-sided (low variance); dev-block errors then showed dirty-lens frames also fail, so the gate became two-sided and the locked run was repeated (`--force`, both logged; the first gave 1.13 cm at 86% coverage). The locked result is therefore not a pristine single look.

Metrics. Mean absolute error (MAE, primary), RMSE and bias (cm), overall and by gauge depth range (dry ≤ 0.5 cm, 0.5–10 cm, >10 cm). Uncertainty is the spread over held-out blocks (95% intervals bootstrap whole blocks; two locked blocks give none, so locked rows show point values).

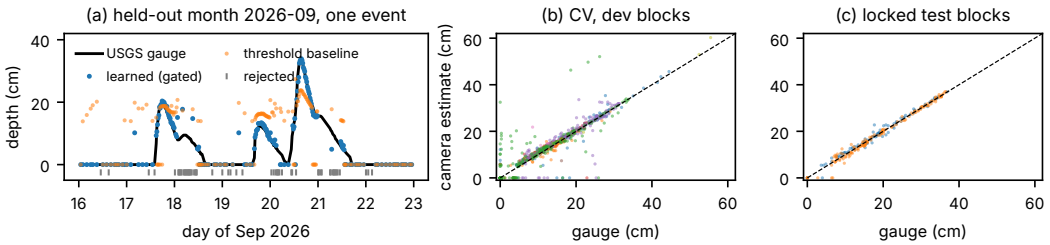


Figure 1: Held-out predictions on real footage. (a) One event in 2026-09 (model never saw this month); grey ticks are rejected frames. (b,c) Camera vs. gauge, gated frames, coloured by month.

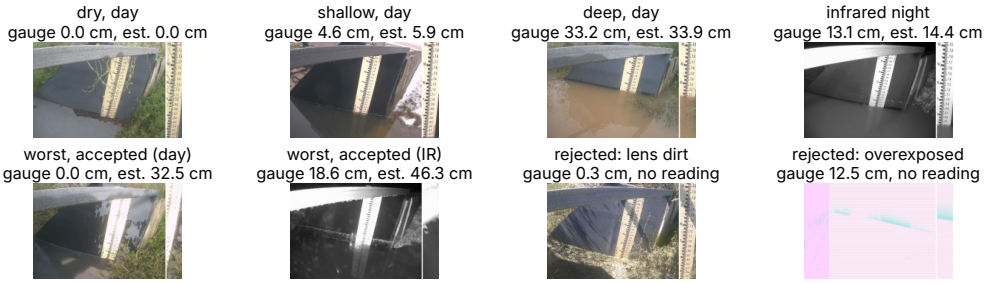


Figure 2: Real frames (scene crop and rectified plate). Top: median-error examples of each category. Bottom: largest accepted errors and rejected frames. Selection rules are fixed in code (file names in `paper/figs/fig_frames_files.json`).

5 Results and Error Analysis

Main result. Table 1 compares the reader with both baselines. On gated frames in CV the learned reader has 1.13 cm MAE (95% CI 0.67–1.69; RMSE 3.1, bias +0.4 cm) versus 6.97 cm for the threshold baseline and 8.80 cm for always-0 cm. On the two locked months it reaches 0.79 cm (baseline 11.52, always-0 18.90). Without the gate the errors are 3.38 cm (CV) and 3.11 cm (locked): the gate buys reliability by abstaining on 33% (CV) and 18% (locked) of frames. Per-month MAE has median 1.06 cm and worst 3.78 cm.

Shallow water. In the 0.5–10 cm range (gated CV, $n = 188$) MAE is 2.0 cm against 6.9 cm for always-0 and 8.0 cm for the baseline (locked: 1.6, 7.6, 7.2; $n = 39$). That is the largest error relative to depth. Depths below the plate bottom (4.3 cm) cannot be observed, and only 7 gated CV frames lie between 0.5 and 3 cm, too few to assess.

Temporal smoothing hurts. A causal median of the last five accepted readings raises MAE from 1.13 to 1.77 cm (CV) and from 0.79 to 2.51 cm (locked): the level moves by tens of centimetres within an hour, so lag outweighs noise reduction. All headline numbers are per-frame.

Failure audit. Fig. 2 (bottom) and the CV subsets show where the error comes from. Infrared frames are *not* worse than daylight (3.30 vs. 3.41 cm). The costly cases are a borrowed pose (9.82 vs. 1.91 cm registered; the camera moved and SIFT failed) and frames outside the gate (7.90 vs. 1.13 cm); high-variance frames (dirt, droplets) average 4.01 cm. In the worst accepted daytime frame (gauge 0 cm) glare washes out the plate and water streaks on the panel behind it look like wet rows; in the worst infrared frame the plate is saturated.

6 Graduate Depth Analysis

Resolution sweep and error model. To vary the scale s, registration is computed once at native resolution (camera set up once), then each frame is area-downsampled by $f \in \{1, \ldots, 64\}$ and the plate re-sampled with $360/f$ rows, giving $s = 0.16f$ cm/px. (Registering at the degraded resolution collapses for $f \geq 8$ and confounds calibration with readout.) The reader is retrained per f with the same CV. Fig. 3a–b: gated MAE is 1.13 cm at the native scale, still 1.36 cm at 2.5 cm/px and

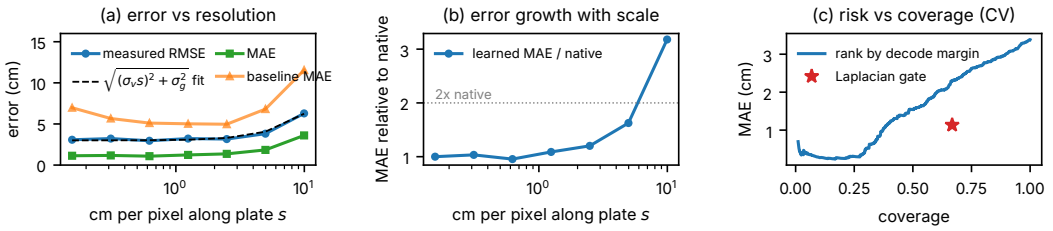


Figure 3: (a) Depth error vs. plate scale s (CV, gated); the dashed line is the fitted error model. (b) Gated MAE relative to its native value. (c) CV MAE vs. coverage when frames are ranked by decode margin; the star is the Laplacian gate.

1.84 cm at 5.0 cm/px, but 3.61 cm at 10.0 cm/px, more than twice native; the baseline goes from 6.97 to 11.55 cm. Fitting $\mathrm{RMSE}^2 = (\sigma_v s)^2 + \sigma_g^2$ to the gated sweep gives $\sigma_v = 0.55$ px and $\sigma_g = 3.0$ cm. A sub-pixel σ_v is plausible because the change point pools hundreds of rows; the σ_g floor dominates at fine scales and reflects gauge offset and residual bad frames. Resolution is therefore not the binding constraint at this camera; the plate bottom and image quality are.

Risk–coverage. Ranking by decode margin lowers MAE as coverage falls (Fig. 3c), but at the gate's coverage it gives 2.28 cm against 1.13 cm for the gate: a cheap image statistic abstains better than the classifier's own confidence, plausibly because pose and lens failures are invisible to a row classifier.

7 Limitations and Conclusion

One camera means the effective sample size is the number of months, not frames, and nothing here shows transfer to other cameras. Labels are a gauge beside the plate, not tape measures. The plate begins at 4.3 cm, so shallower water is unobservable and reported as 0. The locked set is two months, seen twice, and the gate was designed with dev-month error in view. Stronger claims need more labelled cameras and tape-measured depths.

Conclusion. On real USGS footage a learned plate reader with a quality gate reads depth to about 1.13 cm (CV) on accepted frames, far better than a colour-threshold baseline, and stays within twice its native error up to about 5.0 cm/px; it abstains on 33% of frames and cannot observe depths below 4.3 cm.

8 Reproducibility and AI-Use Disclosure

Code, splits, gauge tables, the trained model and all results are in `github.com/kalpshah18/FloodCam` (branch `main`); `scripts/make_paper_assets.py` regenerates every number, table and figure here from `results/`. Frames are public domain; no credentials or private data are used. **AI use:** Claude Code (Anthropic) helped with brainstorming, code, debugging, experiments, the dashboard and drafting this text. I checked the pipeline with unit tests, by inspecting registration strips and failure frames, and by confirming locked data never enter training; I am responsible for all results.

References

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